

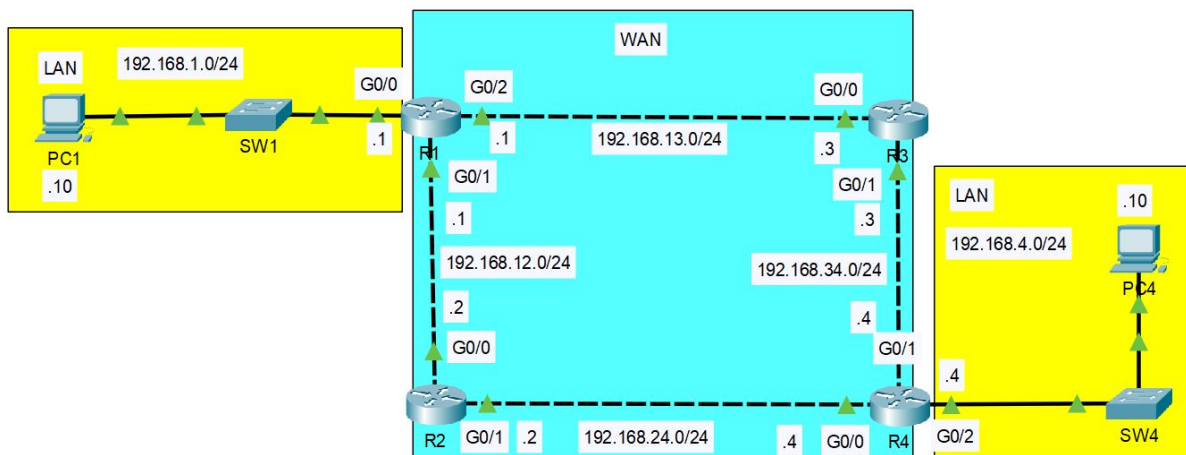
DAY 11 - Routing Fundamental

Routing Fundamental

What is Routing

1. Routing is the process that routers use to determine the path that IP packets should take over a network to reach their destination.
 1. Routers store routes to all of their known destinations in a **Routing Table**.
 2. When routers receive packets, they look in their **Routing Table** to find the best route to forward that packet.
 2. There are *two* main routing methods (methods that routers use to learn routes):
 1. **Dynamic Routing**: Routers use Dynamic Routing Protocols (eg. OSPF) to share routing information with each other automatically and build their routing tables.
 2. **Static Routing**: A network engineer/admin manually configures routes on the router.
 3. A route tells the router: *To send a packet to Destination x , it should send the packet to Next Hop y . (Next Hop = Next Router)*
 1. Or, if the destination is directly connected to the router, send the packet directly to the **Destination**.
 2. Or, if the destination is the router's own IP address, then **Receive the Packet for Yourself** (and don't forward).
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Example Topology:



Pre-routing setup:

- Example:
 - R1> en
 - R1# conf t
 - R1(config)# interface g0/0
 - R1(config-if)# ip address 192.168.1.1 255.255.255.0
 - R1(config-if)# no shutdown
- Repeat for all PC, Routers & Interfaces.

• R1:

```
R1#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	192.168.1.1	YES	NVRAM	up	up
GigabitEthernet0/1	192.168.12.1	YES	NVRAM	up	up
GigabitEthernet0/2	192.168.13.1	YES	NVRAM	up	up

• R2:

```
R2#show ip int br
```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	192.168.12.2	YES	NVRAM	up	up
GigabitEthernet0/1	192.168.24.2	YES	NVRAM	up	up
GigabitEthernet0/2	unassigned	YES	NVRAM	administratively down	down

• R3:

```
Interface
```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	192.168.13.3	YES	manual	up	up
GigabitEthernet0/1	192.168.34.3	YES	manual	up	up
GigabitEthernet0/2	unassigned	YES	unset	administratively down	down

• R4:

```
R4#show ip int brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	192.168.24.4	YES	manual	up	up
GigabitEthernet0/1	192.168.34.4	YES	manual	up	up
GigabitEthernet0/2	192.168.4.4	YES	manual	up	up

View Routing Table (R1) using `show ip route` :

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

Gateway of last resort is not set

```

192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.1.0/24 is directly connected, GigabitEthernet0/0
L    192.168.1.1/32 is directly connected, GigabitEthernet0/0
192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.12.0/24 is directly connected, GigabitEthernet0/1
L    192.168.12.1/32 is directly connected, GigabitEthernet0/1
192.168.13.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.13.0/24 is directly connected, GigabitEthernet0/2
L    192.168.13.1/32 is directly connected, GigabitEthernet0/2
```

Dissecting routing table (R1):

1. Codes:

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

The **Codes** section is the legend for the table which lists the different protocols which routers can use to learn routes.

2. Routes:

We automatically get **2 Routes** without even configuring anything.

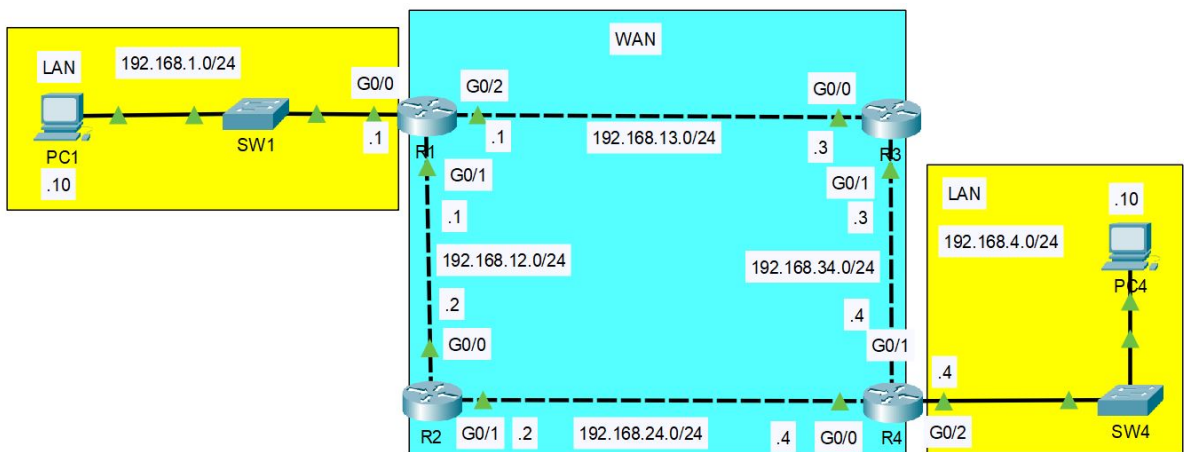
- **L : Local** route, a route to the **Actual IP Address configured on the interface** (With a /32 netmask).

```
R1#show ip int brief
Interface                IP-Address
GigabitEthernet0/0       192.168.1.1
GigabitEthernet0/1       192.168.12.1
GigabitEthernet0/2       192.168.13.1
```

```

L    192.168.1.1/32 is directly connected, GigabitEthernet0/0
192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.12.0/24 is directly connected, GigabitEthernet0/1
L    192.168.12.1/32 is directly connected, GigabitEthernet0/1
192.168.13.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.13.0/24 is directly connected, GigabitEthernet0/2
L    192.168.13.1/32 is directly connected, GigabitEthernet0/2
```

- **C : Connected** route, a route to the **Network ID the interface is connected to** (with the actual netmask configured on the interface.)

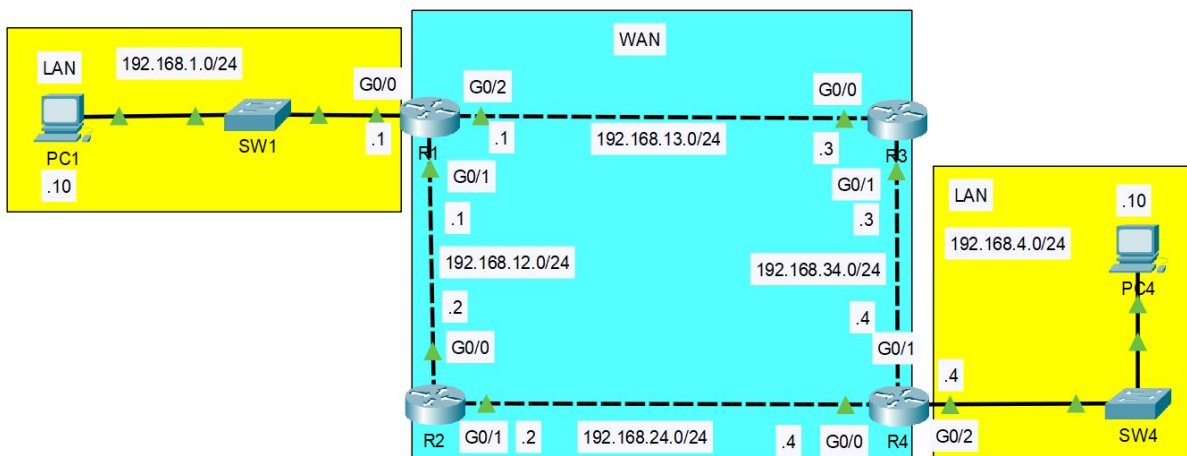


```

192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.1.0/24 is directly connected, GigabitEthernet0/0
L 192.168.1.1/32 is directly connected, GigabitEthernet0/0
192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.12.0/24 is directly connected, GigabitEthernet0/1
L 192.168.12.1/32 is directly connected, GigabitEthernet0/1
192.168.13.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.13.0/24 is directly connected, GigabitEthernet0/2
L 192.168.13.1/32 is directly connected, GigabitEthernet0/2

```

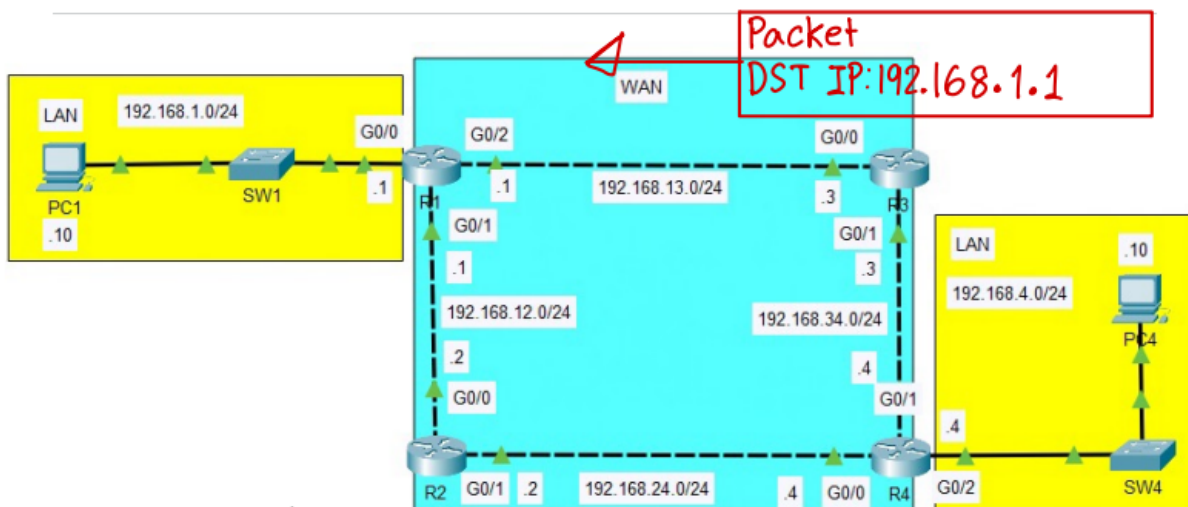
Connected vs. Local Routes:



- A **Connected Route** is a route to the network the *interface* is connected to:
 - **G0/2 of R1:** 192.168.1. 1 /24
 - **Network Address:** 192.168.1. 0 /24
 - It provides a route to **all hosts in that network:**
 - 192.168.1. 3
 - 192.168.1. 135
 - 192.168.1. 254 etc.
 - **R1 knows:**
 - "If I need to send a packet to any host in the 192.168.1.0/24 network, I should send it out using the **G0/2 Interface**"

- 192.168.1. 87 = Within that network, send packet out of **G0/2**
- 192.168. 2 .1 = **Not Within the same Network**, Send out via a different interface or drop the packet.
- A **Local Route** is a route to the *exact* IP address configured on the interface.
 - A /32 netmask is used to specify the exact IP address of the interface.
 - **R1** knows:
 - "If I receive a packet destined to this IP address, the message *is for me*"

Route Selection:



```

192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.1.0/24 is directly connected, GigabitEthernet0/0
L    192.168.1.1/32 is directly connected, GigabitEthernet0/0

```

Imagine a packet with **Destination**: 192.168.1.1 coming from **G0/0 R3**.

- The packet is matched by two routes:
 - C route: 192.168.0/24 for the network IP.
 - L route: 192.168.1.1/32 for the R1's G0/0 IP.
- Which route will R1 use?
 - It will choose the **MOST SPECIFIC Matching Route**.
 - The route to 192.168.1.0/24 includes **256** different IP addresses (192.168.1.1 to 192.168.1.255)
 - The route to 192.168.1.1/32 includes **1** IP address (only 192.168.1.1)
 - We can see that 192.168.1.1/32 route is the most specific.

- Thus, the route **R1** uses is `L 192.168.1.1/32` .

Misc.

```

192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.1.0/24 is directly connected, GigabitEthernet0/0
L    192.168.1.1/32 is directly connected, GigabitEthernet0/0
192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.12.0/24 is directly connected, GigabitEthernet0/1
L    192.168.12.1/32 is directly connected, GigabitEthernet0/1
192.168.13.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.13.0/24 is directly connected, GigabitEthernet0/2
L    192.168.13.1/32 is directly connected, GigabitEthernet0/2

```

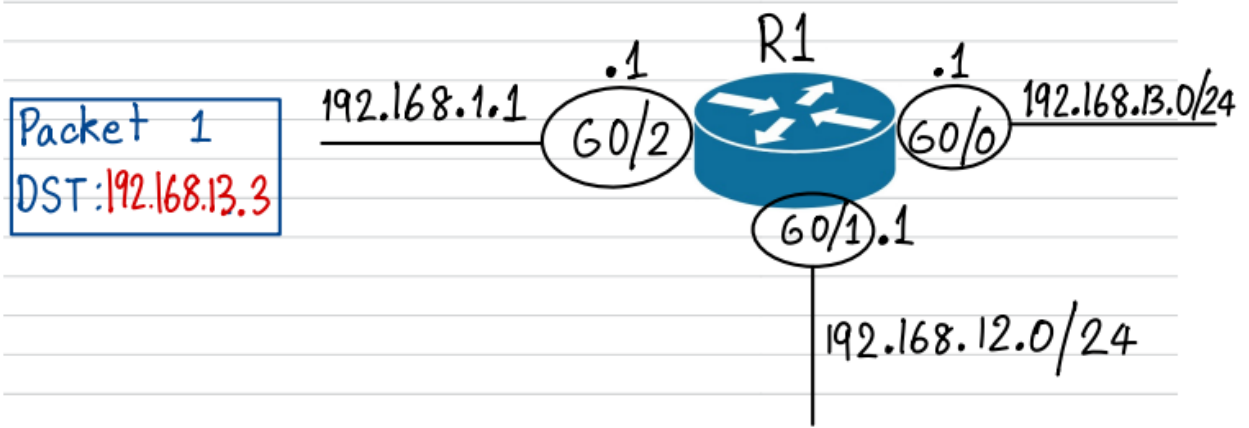
These three lines are not routes, they mean:

1. **192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks**
 1. In the routing table, there are **Two Routes** to subnets that fit within the `192.168.1.0/24` class C network, with **two different netmasks** (`24` and `32`)
2. **192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks**
 1. In the routing table, there are **Two Routes** to subnets that fit within the `192.168.12.0/24` class C network, with **two different netmasks** (`24` and `32`)
3. **192.168.13.0/24 is variably subnetted, 2 subnets, 2 masks**
 1. In the routing table, there are **Two Routes** to subnets that fit within the `192.168.13.0/24` class C network, with **two different netmasks** (`24` and `32`)

We will cover **Subnetting** later, for now just know that these three lines are not routes.

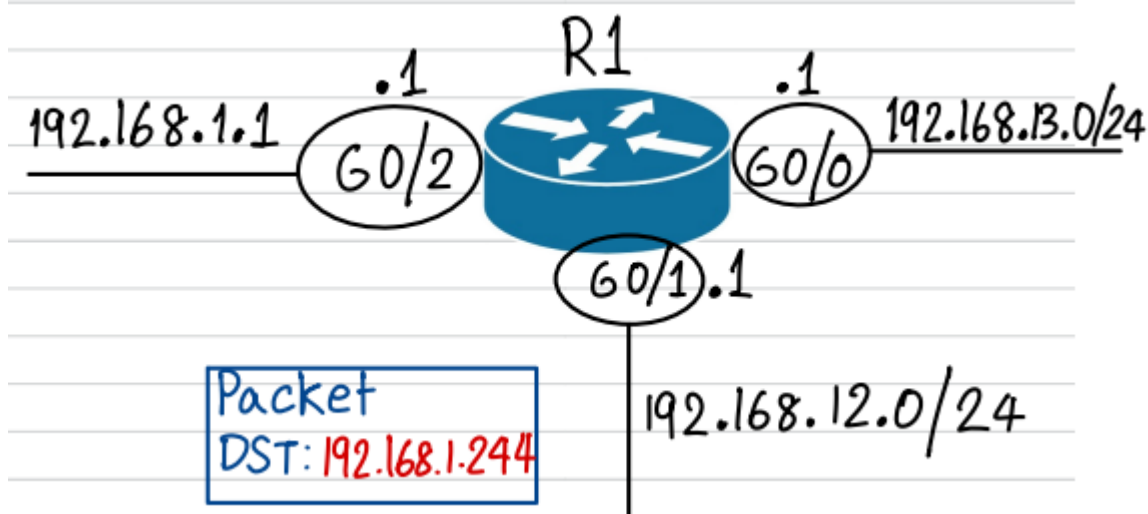
Route Selection (Practice):

1. Packet 1:



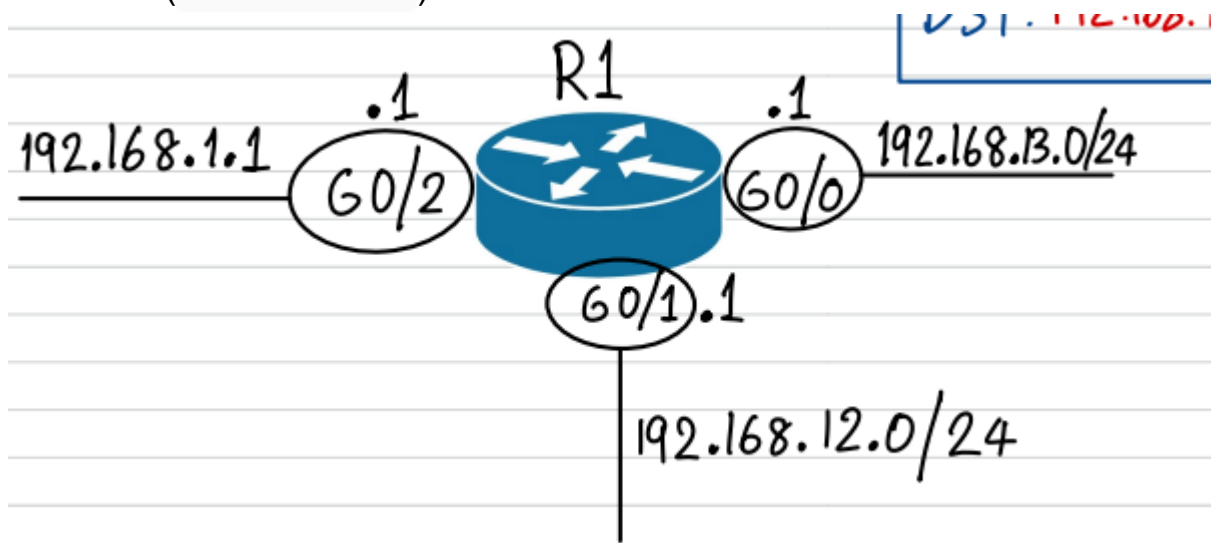
- Most matching route: 192.168.13.0/24 via **G0/0** interface.

2. Packet 2:



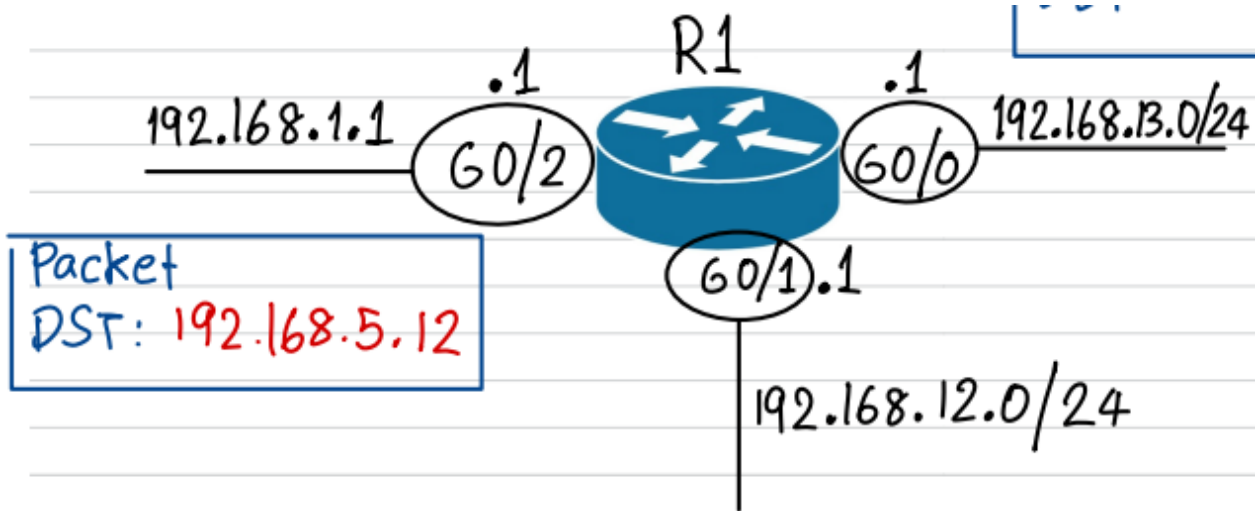
- Most matching route: 192.168.1.0/24 via **G0/2** interface.

3. Packet 3 (192.168.12.1):



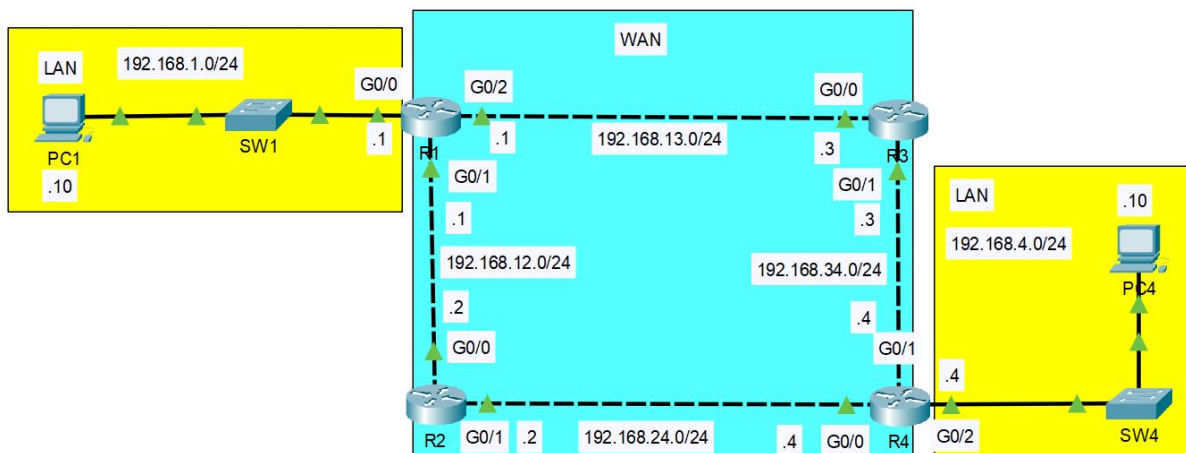
- Most matching route: 192.168.12.1/32 At the **G0/1** interface.

4. Packet 4:



- Most matching route: **None**, R1 will drop the packet.

Summary



- Routers store information about destinations they know in their *Routing Table*.
 - When they receive packets, they look in the routing table to find the best route to forward the packet.
- Each **Route** in the routing table is an instruction:
 - To reach destinations in **Network X**, send the packet to **Next-Hop Y** (The next router in the path to the destination).
 - If the destination is directly connected (**Connected Route**), then send the packet directly to the destination.
 - If the destination is your own IP Address (**Local Route**), then receive the packet for yourself.
- When you configured an IP address on an interface and enable the interface, **Two Routes** are automatically added to the routing table:

- **Connected Route** (Code **C** in the routing table): A route to the network connected to the interface.
 - If the interface's IP is `192.168.1.1/24`, the **C** route will be to `192.168.1.0/24`
 - Tells the router: "To send a packet to a destination in this network, send it out of the interface specified in the route."
 - **Local Route** (Code **L** in the routing table): A route to the **exact** IP address configured on the interface.
 - If the interface's IP is `192.168.1.1/24`, the **L** route will be to `192.168.1.1/32`
 - Tells the router: "Packets to this destination are for you, you should receive them for yourself only".
 - A route **Matches** a destination if the packet's destination IP address is part of the network specified in the route.
 - A packet to `192.168.1.70` is matched by a route to `192.168.1.0/24` **but not** `192.168.0.0/24`
 - If a router receives a packet and it doesn't have a route that matches the packet's destination, it will **Drop** the packet.
 - This is different from **Switches**, which will **Flood** the frames if they don't have a **MAC Entry** for the Frame's destination.
 - If a router receives a packet and it has multiple routes that match the packet's destination, it will use the **Most Specific** matching route to forward the packet.
 - **Most Specific Matching Route** = The matching route with the **longest** prefix length.
 - `192.168.1.1 : 192.168.1.1/32 > 192.168.1.0/24`
 - This is different from **Switches**, which look for an **Exact Match**.
-